

Dietary Fiber Future Directions: Integrating New Definitions and Findings to Inform Nutrition Research and Communication



The CODEX Alimentarius definition of dietary fiber includes all nondigestible carbohydrate polymers with a degree of polymerization of 3 or more as dietary fiber with the proviso that they show health benefits. The global definition, if accepted by all authoritative bodies, offers a chance for international harmonization in research, food composition tables, and food labeling. Its nonacceptance highlights problems that may develop when definitions vary by region. The definition requires that the research community agrees upon physiological effects for which there is substantial scientific agreement, e.g., fibers' effects on laxation and gut health, on attenuating blood lipids and blood glucose and insulin, and in promoting fermentation in the large bowel. The definition also necessitates the delineation of research protocols to prove the benefits of various isolated and synthesized fibers. These should emanate from evidence-based reviews that fairly weigh epidemiological data while considering that added fibers are not reflected in many food composition databases. They then should include well-controlled, randomized, control trials and utilize animal studies to determine mechanisms. Agreement on many study variables such as the type of subject and the type of baseline diet that best fits the question under investigation will also be needed. Finally, the definition establishes that all types of fiber can address the severe fiber consumption gap that exists throughout the world by recognizing that the combination of fiber-rich and -fortified foods increases fiber intake while allowing consumers to stay within allowed energy levels.